

Problem 1. Find the (slant) surface area of a right circular cone of height 2cm and a base radius of 2cm using the Cartesian as well as the Spherical coordinates. The answers should be the same, equal to $4\pi\sqrt{2}\text{cm}^2$.

Try the same using Cylindrical coordinates, and you will notice that it gets a little trickier!

Problem 2. Using double integrals and the **Cylindrical coordinates**, compute the surface area of a cylinder of radius 2, with $0 \leq z \leq 2$.

Then compute the total surface area of the wedge shape formed by cutting the cylinder above with two half-planes $\theta = \frac{\pi}{6}$ and $\theta = \frac{\pi}{3}$, and having the top and bottom surfaces ($z = 0$ and $z = 2$ respectively) closed as well.

Problem 3. Using the **Spherical coordinates**, compute the surface area of a sphere of radius $\rho = R$. Your answer should be $4\pi R^2$.

Problem 4. Use **Spherical coordinates** to find the surface area of the cone $\phi = \frac{\pi}{6}$ with $0 \leq \rho \leq 2$. Your answer should be the same as what you get from the formula $\pi \cdot r \cdot \ell$, where r is the base radius of the cone, and ℓ is the slant height.

Problem 5. By integrating surface element of a sphere, find area of a strip on the surface of a sphere from $\theta = 45$ degrees to $\theta = 135$ degrees (surface radius is R).

Problem 6. By integrating all three surface elements of a cylinder, find the surface area of a cylindrical pie:

$$R/2 \leq \rho \leq R,$$

$$\pi/2 \leq \phi \leq 3\pi/4,$$

$$1 \leq z \leq 3.$$

Suggestion: You should try to draw/sketch each of these surfaces in problems 1 - 6 before attempting the solutions.